

Atomic Structure

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Rutherford's atomic model

- (1) An atom consists of a small nucleus containing all positive charges of the atom and practically the whole of its mass
- (2) The nucleus is surrounded by a number of electrons equal to the number of positive charges on the nucleus.
- (3) The electrons are in constant motion round the nucleus like that of planets round the sun in such a way that the electrostatic force of attraction between the electrons and positive nucleus is counterbalanced by the centrifugal force.

Limitations:

- 1) Rutherford's model could not explain the stability of atom. According to electromagnetic theory, the charged particle under acceleration should continuously emit radiation. Hence the electron moving in the orbits must lose energy and fall into the nucleus. But this is not happening.
- 2) This model could not explain the electronic structure and energy of electrons.
- 3) It could not explain the atomic spectra.

Bohr's Model

In 1913, Niels Bohr (1885-1962) proposed another model of the atom where electrons move around the nucleus in circular paths. Bohr's atomic model is built upon a set of postulates, which are as follows :

1. The electrons move in a definite circular paths around the nucleus . He called these circular paths as orbits and postulated that as long as the electron is in a given orbit its energy does not change (or energy remains fixed). These orbits were therefore referred to as stationary orbits or stationary states or non radiating orbits.
2. The electron can change its orbit by absorbing or releasing energy. An electron at a lower (initial) state of energy, E_i can go to a (final) higher state of energy, E_f by absorbing a single photon of energy given by

$$E = h\nu = E_f - E_i$$

Similarly, when electron changes its orbit from a higher initial state of energy E_i to a lower final state of energy E_f , a single photon of energy $h\nu$ is released.

3. The angular momentum of an electron of mass m_e moving in a circular orbit of radius r and velocity v is an integral multiple of $\mathbf{h}/2$.

$$\mathbf{mvr} = \mathbf{nh}/2$$

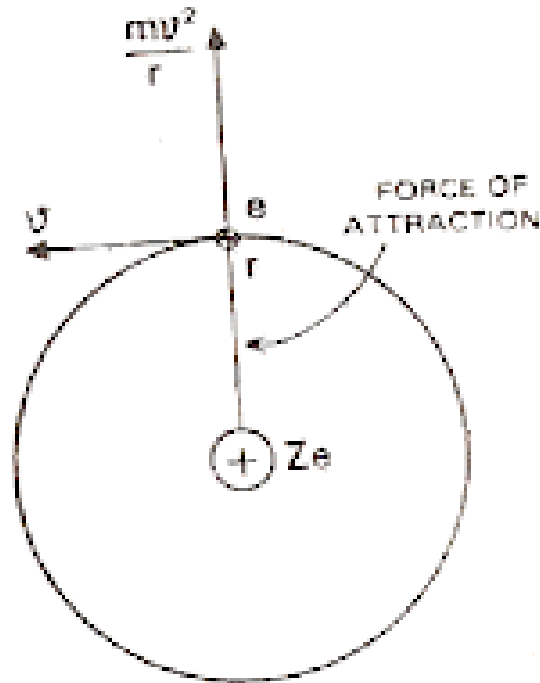
where n is a positive integer, known as the principal quantum number.

Limitations:

- (1) Bohr's model of an atom failed to explain the Zeeman Effect (effect of magnetic field on the spectra of atoms).
- (2) It also failed to explain the Stark effect (effect of electric field on the spectra of atoms).
- (3) It could not explain the spectra obtained from larger atoms.

Calculation of radius of orbits:

Consider an electron of charge e revolving around a nucleus of charge Ze , where Z is the atomic number and e the charge on a proton. Let m be the mass of the electron, r the radius of the orbit and v the tangential velocity of the revolving electron. The electrostatic force of attraction between the nucleus and the electron (Coulomb's law),



Consider an electron of charge 'e' revolving around a nucleus of charge 'ze', where 'z' is the atomic number and 'e' the charge on a proton. Let 'm' be the mass of the electron, 'r' be the radius of the orbit and 'v' the tangential velocity of the revolving electron.

The electrostatic force of attraction between the nucleus and the electron (coulomb's law) = $ze \times e / r^2$.

The centrifugal force acting on the electron = mv^2 / r

Bohr assumed that these two opposing forces must be balancing each other exactly to keep the electron in orbit. Thus,

$$Ze^2 / r^2 = mv^2 / r$$

For hydrogen $z = 1$, therefore,

$$e^2 / r^2 = mv^2 / r \text{ -----(1)}$$

Multiplying both sides by 'r'

$$e^2 / r = mv^2 \text{ -----(2)}$$

According to one of the postulates of the Bohr's theory, angular momentum of the revolving electron is integral multiple of $h/2\pi$.

$$mvr = nh/2\pi$$

$$\text{or, } v = nh/2\pi mr \text{-----(3)}$$

Substituting the value of 'v' in equation (2),

$$e^2/r = m(nh/2\pi mr)^2$$

$$\text{So, } r = n^2h^2/4\pi^2me^2 \text{-----(4)}$$

Since the value of h, m and e had been determined experimentally, substituting these values in (4), we have $r = n^2 \times 0.529 \times 10^{-8} \text{ cm} \dots(5)$

where n is the principal quantum number and hence the number of the orbit. When $n = 1$, the equation (5) becomes

$$r = 0.529 \times 10^{-8} \text{ cm} = \alpha_0 \dots(6)$$

This last quantity, α_0 called the first Bohr radius was taken by Bohr to be the radius of the hydrogen atom in the ground state. This value is reasonably consistent with other information on the size of atoms. When $n = 2, 3, 4$ etc., the value of the second and third orbits of hydrogen comprising the electron in the excited state can be calculated.

Energy of electron in each orbit:

For hydrogen atom, the energy of the revolving electron, E is the sum of its kinetic energy ($\frac{1}{2} mv^2$) and potential energy ($-\frac{e^2}{r}$)

$$E = \frac{1}{2} mv^2 - \frac{e^2}{r} \dots \dots \dots (7)$$

From equation (1)

$$mv^2/r = e^2/r$$

Substituting the value of mv^2 in (7)

$$E = \frac{1}{2} \times \frac{e^2}{r} - \frac{e^2}{r}$$

$$\text{or, } E = -\frac{e^2}{2r} \dots \dots \dots (8)$$

Substituting the value of r from equation (4) in (8)

$$\begin{aligned} E &= -\frac{e^2}{2} \times \frac{4\pi^2 m^2}{n^2 h^2} \\ &= -\frac{2\pi^2 m e^4}{n^2 h^2} \end{aligned}$$

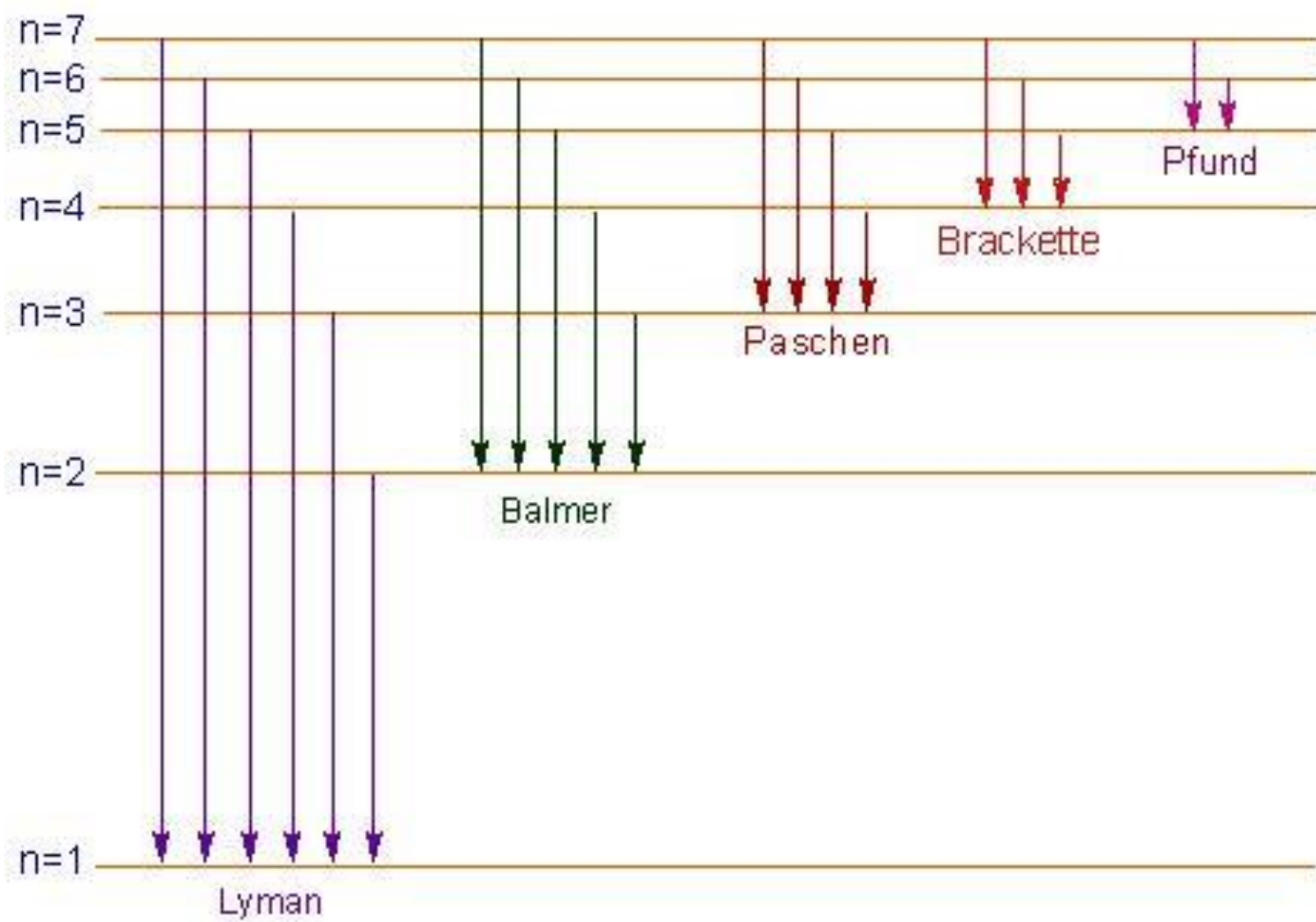
Substituting the value of r from equation (4) in (8)

$$\begin{aligned} E &= -\frac{e^2}{2} \times \frac{4\pi^2 m^2}{n^2 h^2} \\ &= -2\pi^2 m e^4 / n^2 h^2 \end{aligned}$$

Explanation of Line Spectrum of Hydrogen Atom

When a high potential is applied to hydrogen gas at low pressure in a discharge tube, it starts emitting a bright light. It is separated into several radiations and forms a spectrum upon passing through a prism or grating. The spectrum consists of separate lines corresponding to different wavelengths. This is called Hydrogen atomic spectrum. The spectral lines are formed due to electronic transitions from one energy level to another. These lines are divided into five series according to the range of wavelengths as follows.

Spectral series	Spectral region	n_1	n_2
1. Lyman series	Ultra-violet	1	2,3,4,5,6,7, _ _ _ _
2. Balmer series	Visible	2	3,4,5,6,7, _ _ _ _
3. Paschen series	near infra-red	3	4,5,6,7, _ _ _ _
4. Brackett series	infra-red	4	5,6,7, _ _ _ _
5. Pfund series	far infra-red	5	6,7, _ _ _ _



The wave numbers of spectral lines in each series can be calculated using Rydberg's equation as follows

$$\bar{\nu} = R_H Z^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

Where n_1 and n_2 are the principal quantum numbers of orbits corresponding to the electronic transition. R_H = Rydberg's constant = $1,09,677 \text{ cm}^{-1}$, Z = atomic number.

$$R_H = \text{Rydberg constant} = \frac{m_e e^4}{8 \epsilon_0^2 h^3 c} = 1.0973 \times 10^7 \text{ m}^{-1}$$

$$m_e = \text{mass of electron} = 9.11 \times 10^{-31} \text{ kg}$$

$$e = \text{charge of electron} = 1.602 \times 10^{-19} \text{ C}$$

$$\epsilon_0 = \text{vacuum permittivity} = 8.85 \times 10^{-12} \text{ C}^2 \text{N}^{-1} \text{m}^{-2}$$

$$h = \text{Planck's constant} = 6.626 \times 10^{-34} \text{ Js}$$

$$c = \text{speed of light} = 2.998 \times 10^8 \text{ ms}^{-1}$$

Electronic structure of Atom

The new quantum mechanics, which takes into account the particulate and wave nature of matter, is termed the Wave mechanics.

de Broglie's Equation

de Broglie had arrived at his hypothesis with the help of Planck's Quantum Theory and Einstein's Theory of Relativity. He derived a relationship between the magnitude of the wavelength associated with the mass 'm' of a moving body and its velocity. According to Planck, the photon energy 'E' is given by the equation

$$E = h\nu \dots(i)$$

where h is Planck's constant and ν the frequency of radiation. By applying Einstein's mass-energy relationship, the energy associated with photon of mass 'm' is given as

$$E = mc^2 \dots(ii)$$

where c is the velocity of radiation

Comparing equations (i) and (ii)

$$mc^2 = h\nu = h c/\lambda \quad (\because \nu = c/\lambda)$$

$$\text{or, } mc = h/\lambda \dots\dots\text{(iii)}$$

$$\text{mass} \times \text{velocity} = h/\text{wavelength}$$

$$\text{or, momentum (p)} = h/\text{wavelength}$$

$$\text{or, momentum} \propto 1/\text{wavelength}$$

The equation (iii) is called de Broglie's equation and may be put in words as : The momentum of a particle in motion is inversely proportional to wavelength, Planck's constant 'h' being the constant of proportionality.

Dual Nature of Electrons

French physicist named Louis de-Broglie in 1924 said electrons have dual nature, as both particle and wave nature can be seen in an electron.

It stated that when small particles like electrons can pass wave nature when they are in motion, it was a new theory.

Now, de-Broglie introduced the wavelength

$$\lambda = h/mv \dots \dots \dots (1)$$

Where, m is mass of particle, v is velocity and h is planck's constant.

Derivation of de-Broglie's equation:

According to planck's equation

$$E = hf = hc/\lambda \dots (2)$$

Where, E is energy, f is frequency and c is speed of light.

Now, according to Einstein's mass energy relationship

$$hc/\lambda = mc^2$$

$$\text{or, } hc = mc^2\lambda$$

Change the above equation with respect to λ

$$\lambda = hc/mc^2$$

Cancel c from numerator and denominator

$$\lambda = h/mc \quad (4)$$

This is the same as de-Broglie's equation.

Also, Davison and Germer, by performing diffraction effects on an electron beam, verified the dual nature of the electron. Consider an electron accelerated at potential V , then its kinetic energy is

$$\frac{1}{2}(mv^2) = eV$$

$$\text{or, } mV^2 = 2eV$$

$$\text{or, } mV = 2e$$

Now, by multiplying mV on both the sides

$$(mV)^2 = 2eVm$$

taking $\sqrt{\quad}$ on both the sides

$$mV = \sqrt{2eVm} \quad \text{---(5)}$$

Substituting equation (5) and (1) we get

$$\lambda = h / \sqrt{2eVm} \quad \text{---(6)}$$

Now, Bohr's theory can also be related to de-Broglie's equation, therefore wavelength of an electron is calculated by Bohr's orbit and can be related to its circumference that is $2\pi r$, and by multiplying with a whole number n .

$$2\pi r = n\lambda$$

Now, from de-Broglie's equation, by applying

$$2\pi r = nh/mV$$

$$\text{Then, } h/mV = 2\pi r/n$$

Now, by interchanging quantity we get

$$mVr = nh/2\pi \text{---(7)}$$

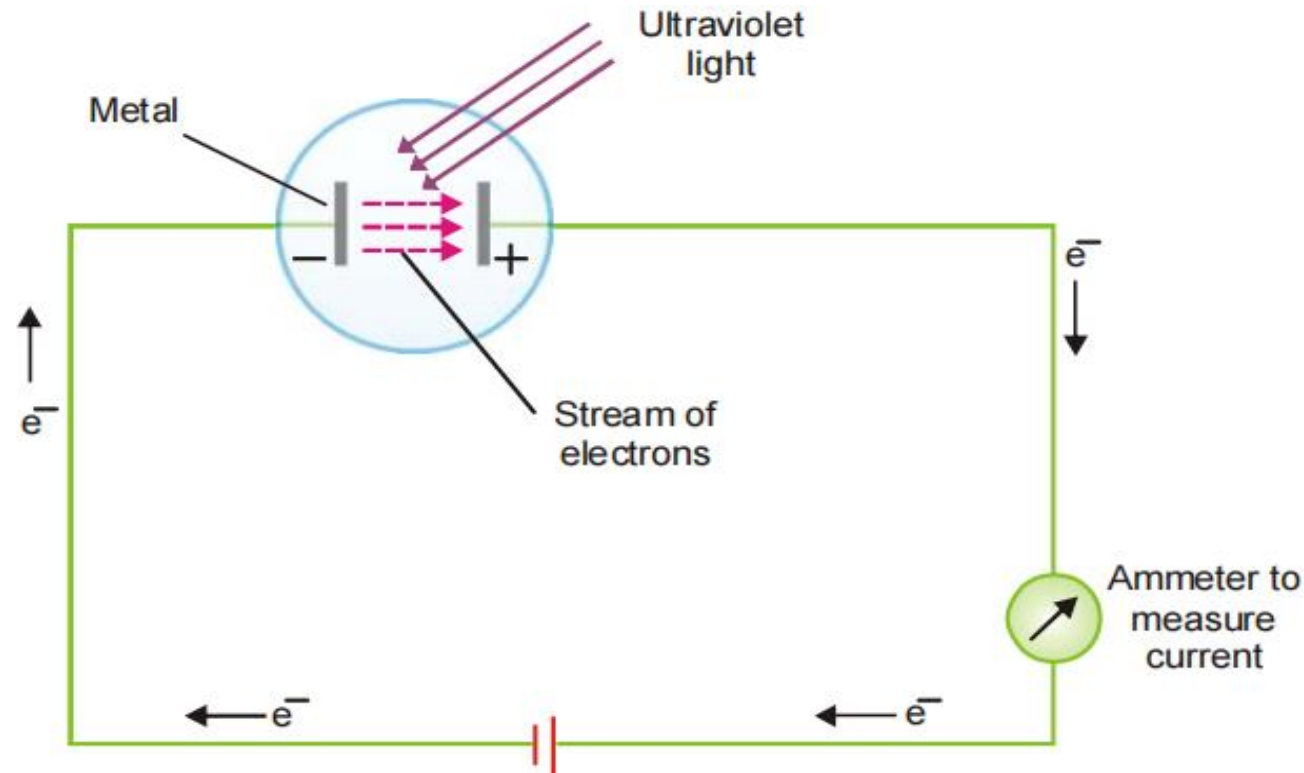
Which is considered as Bohr's equation.

Now, de-Broglie is not significant in daily life because it is considered for micro-particle and not on macro-particle.

[Dual nature of electrons can be related to dual nature of light. Light behaves as both particles and waves. It acts as an electromagnetic wave when in transmission in space, and interacting with electrons acts as if a particle i.e. photon.]

Photoelectric Effect

When a beam of light of sufficiently high frequency is allowed to strike a metal surface in vacuum, electrons are ejected from the metal surface. This phenomenon is known as Photoelectric effect and the ejected electrons Photoelectrons. For example, when ultraviolet light shines on Cs(or Li, Na, K, Rb) as in the apparatus shown in Fig , the photoelectric effect occurs.



With the help of this photoelectric apparatus the following observations can be made :

- (1) An increase in the intensity of incident light does not increase the energy of the photoelectrons.
- (2) The kinetic energy of the photoelectrons increases linearly with the frequency of the incident light.

Quantum number

Quantum numbers are used to describe the position and energy of an electron in an atom. There are four types of quantum numbers.

- i) Principal quantum number
- ii) Azimuthal quantum number
- iii) Magnetic quantum number
- iv) Spin quantum number

Principal quantum number

1. Principal quantum number was proposed by Bohr to explain the hydrogen atomic spectrum.
2. It denotes the main energy level (or shell or orbit).
3. It is denoted by 'n'. It can have positive integral values from 1,2,3,4,____ (or K,L,M,N____).
4. It also describes the energy and size of the orbit. The energy and the size of orbit increases with increase in 'n' value.
5. The maximum number of electrons that can be accommodated in a given shell is equal to $2n^2$.

Azimuthal quantum number

1. Azimuthal quantum number was proposed by Sommerfield in order to explain the fine hydrogen atomic spectrum.
2. It denotes the sub-levels (or sub-shells) in the atom. It is denoted by 'l'
3. The number of sub-levels in a given main energy level is equal to the principal quantum number 'n'. These sub-levels are indicated by azimuthal quantum number 'l' which can have values from 0 to n-1. For ex: In an orbit of principal quantum number n=3, there are three sublevels denoted by $l = 0, 1$ and 2.
4. Azimuthal quantum number also describes the angular momentum of electron and shape of orbitals present in that sublevel.

E.g. $l = 0$ (s)--- spherical

$l = 1$ (p)--- dumbbell

$l = 2$ (d)--- double dumbbell

Magnetic quantum number

1. Magnetic quantum number was proposed by Lande in order to explain the Zeeman and Stark effects. The splitting of spectral lines in strong magnetic field is called Zeeman effect and splitting in strong electric field is called Stark effect.

2. It is denoted by 'm'. It describes the orientation of orbitals.

3. The number of orientations possible for orbitals in a given sublevel 'l' is equal to the $(2l+1)$ and can have values from $-l$ to $+l$.

E.g. The number of orientations (m values) possible for a sublevel $l=2$ is $(2 \times 2) + 1 = 5$ and the values of $m = -2, -1, 0, +1$ and $+2$.

Spin quantum number

1. It was proposed by Goudsmit and Uhlenbeck to explain the double line structure of alkali metal spectra.
2. Spin quantum number denotes the spin of the electron on its own axis. It is denoted by 's'. It can have only two values($+1/2$ and $-1/2$).
3. The clockwise spin is indicated by $+1/2$ or upwards arrow($\uparrow$) and anti-clockwise spin by $-1/2$ or downwards arrow($\downarrow$)

Aufbau (or building up) Principle

This principle is concerned with the energy of the atom and states that the electrons should occupy the orbitals in such a way that the energy of atom is minimum. In other words the electrons in an atom are filled in the increasing order of their energies

i.e. the orbital with the lowest energy is filled up first while that with the highest energy is filled up in the end.

The order of orbital energies can be determined by the following $(n + l)$ rules

Rule 1: An orbital with a lower value for $(n + l)$ has lower energy. For example, the 4s orbital ($n + l = 4 + 0 = 4$) will fill before a 3d orbital ($n + l = 3 + 2 = 5$).

Rule 2: If the value of $(n + l)$ is same for two orbitals then the orbital with lower value of n will be filled first. For example, the 3d orbital ($n + l = 3 + 2 = 5$) will fill before a 4p orbital ($n + l = 4 + 1 = 5$). Following these rules the increasing order of the orbital energies comes out to be

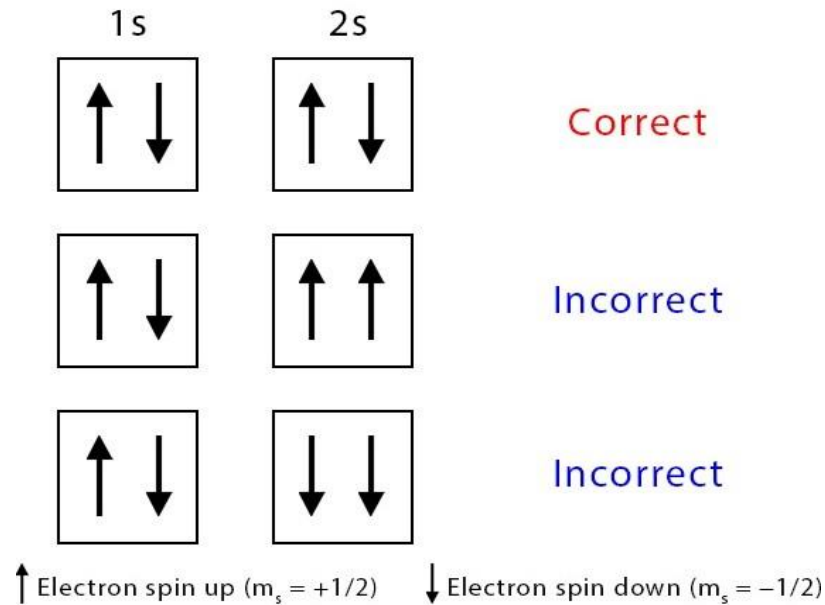
$$1s < 2s < 2p < 3s < 3p < 4s < 3d < 4p < 5s < 4d < 5p < 6s$$

Pauli's Exclusion Principle

This principle concerns the spin of electrons present in an orbital.

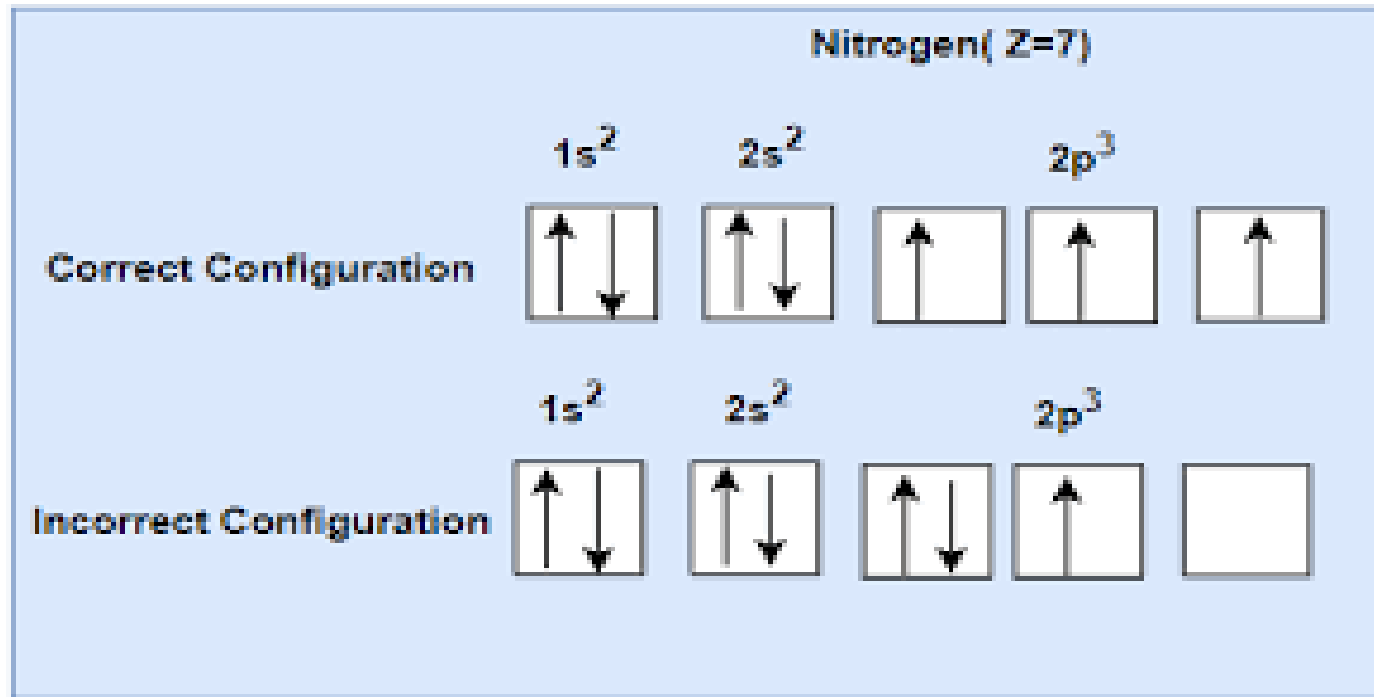
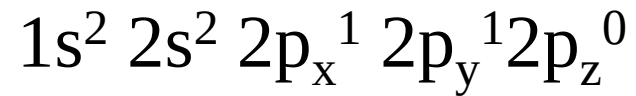
According to the Pauli's principle, **no two electrons can have all the four quantum numbers to be same.**

For example, if a given electron in an atom has the set of four quantum numbers as $n = 2$, $l = 1$, $m_l = 1$ and $m_s = +\frac{1}{2}$ then no other electron in the atom can have the same set of quantum numbers.



Hund's Rule

Every orbital in a subshell is singly occupied with one electron before any one orbital is doubly occupied, and all electrons in singly occupied orbitals have the same spin. For example, the six electrons in carbon distribute as



Heisenberg uncertainty principle

An important consequence of the wave-particle duality of matter and radiation was discovered by Werner Heisenberg in 1927 and is called the **uncertainty principle**.

According to this principle it is not possible to simultaneously measure both the **position** and **momentum (or velocity)** of an electron accurately. In simple words we may state that more accurately you measure a particle's position, the less accurately you're able to measure its momentum, and vice versa. Mathematically, the Heisenberg principle can be expressed in terms of an inequality

$$\Delta x \Delta p \geq h/4\pi$$

Where Δx and Δp are the uncertainties in the measurements of position and momentum respectively. If the position of an object is known exactly (i.e., $\Delta x = 0$), then the uncertainty in the momentum must be infinite, meaning that we cannot say anything about the velocity. Similarly, if the velocity is known exactly, then the position would be entirely uncertain and the particle could be anywhere. It means that we cannot say anything about the position of the particle. In actual practice none of the two properties can be measured with certainty. Due to the small value of the Planck's constant, h (6.626×10^{-34} J s) this principle is not relevant while making measurements of large objects like car, bus or aeroplane etc.

It is relevant, only when you are making measurements on very small objects such as electrons. Heisenberg's principle questioned the validity of Bohr's model. It is so because according to Bohr's model we can precisely calculate the radius of the orbit (i.e., the position of the electron) and the velocity of electron in it. But it is not possible according to Heisenberg's principle. It motivated many scientists to develop newer models of the atom using the dual nature of the electron.

Schrödinger's Wave Equation

In order to provide sense and meaning to the probability approach, Schrödinger derived an equation known after his name as Schrödinger's Wave Equation. Calculation of the probability of finding the electron at various points in an atom was the main problem before Schrödinger. His equation is the keynote of wave mechanics and is based upon the idea of the electron as 'standing wave' around the nucleus.

$$\psi = A \sin 2\pi \frac{x}{\lambda} \quad \dots(a)$$

where ψ (pronounced as *sigh*) is a mathematical function representing the amplitude of wave (called *wave function*) x , the displacement in a given direction, and λ , the wavelength and A is a constant.

By differentiating equation (a) twice with respect to x , we get

$$\frac{d\psi}{dx} = A \frac{2\pi}{\lambda} \cos 2\pi \frac{x}{\lambda} \quad \dots(1)$$

and

$$\frac{d^2\psi}{dx^2} = -A \frac{4\pi^2}{\lambda^2} \sin 2\pi \frac{x}{\lambda} \quad \dots(2)$$

But $A \sin 2\pi \frac{x}{\lambda} = \psi$

$$\therefore \frac{d^2\psi}{dx^2} = -\frac{4\pi^2}{\lambda^2} \psi \quad \dots(3)$$

The K.E. of the particle of mass m and velocity v is given by the relation

$$\text{K.E.} = \frac{1}{2}mv^2 = \frac{1}{2} \frac{m^2 v^2}{m} \quad \dots(4)$$

According to Broglie's equation

$$\lambda = \frac{h}{mv}$$

or $\lambda^2 = \frac{h^2}{m^2 v^2}$

or $m^2 v^2 = \frac{h^2}{\lambda^2}$

Substituting the value of $m^2 v^2$, we have

$$\text{K.E.} = \frac{1}{2} \times \frac{h^2}{m\lambda^2} \quad \dots(5)$$

From equation (3), we have

$$\lambda^2 = -\frac{4\pi^2\psi}{\frac{d^2\psi}{dx^2}} \quad \dots(6)$$

Substituting the value of λ^2 in equation (5)

$$\begin{aligned}\text{K.E.} &= -\frac{1}{2m} \cdot \frac{h^2}{4\pi^2\psi} \cdot \frac{d^2\psi}{dx^2} \\ &= -\frac{h^2}{8\pi^2 m\psi} \cdot \frac{d^2\psi}{dx^2}\end{aligned}$$

The total energy E of a particle is the sum of kinetic energy and the potential energy
i.e.,

$$E = \text{K.E.} + \text{P.E.}$$

or

$$\text{K.E.} = E - \text{P.E.}$$

$$= -\frac{h^2}{8\pi^2 m\psi} \cdot \frac{d^2\psi}{dx^2}$$

or

$$\frac{d^2\psi}{dx^2} = -\frac{8\pi^2 m}{h^2} (E - \text{P.E.}) \psi$$

$$\frac{d^2\psi}{dx^2} + \frac{8\pi^2m}{h^2} (E - \text{P.E.}) \psi = 0$$

This is Schrödinger's equation in one dimension. It need be generalised for a particle whose motion is described by three space coordinates x , y and z . Thus,

$$\frac{d^2\psi}{dx^2} + \frac{d^2\psi}{dy^2} + \frac{d^2\psi}{dz^2} + \frac{8\pi^2m}{h^2} (E - \text{P.E.}) \psi = 0$$

This equation is called the **Schrödinger's Wave Equation**. The first three terms on the left-hand side are represented by $\Delta^2\psi$ (pronounced as del-square sigh).

$$\Delta^2 \psi + \frac{8\pi^2m}{h^2} (E - \text{P.E.}) \psi = 0$$